

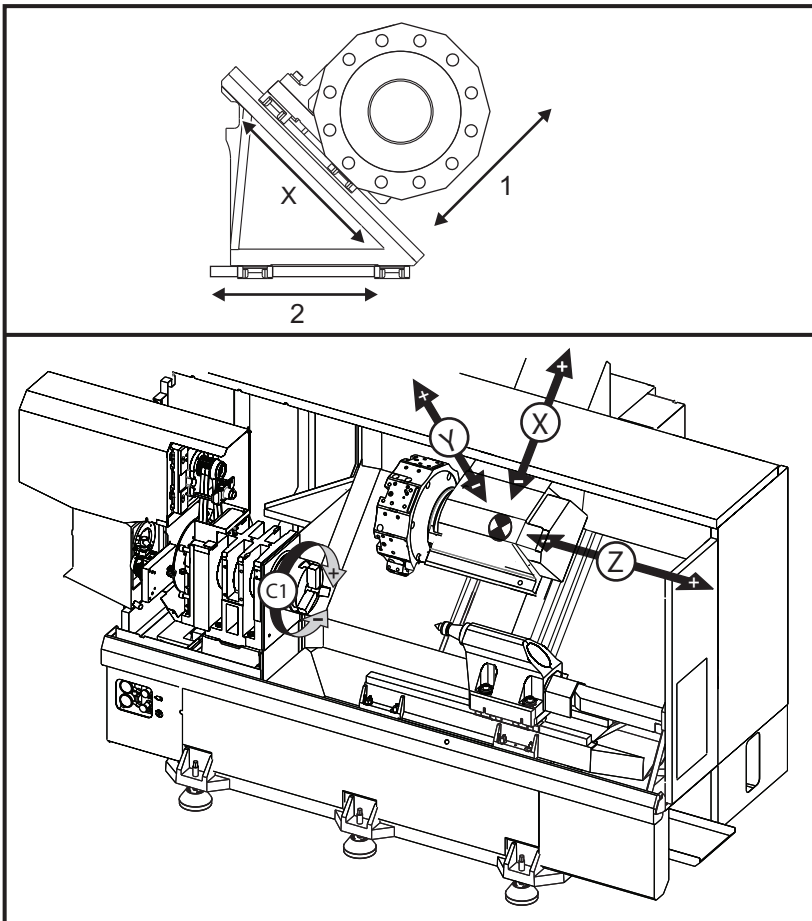
5. Type a value for the highlighted variable and press **[ENTER]**. The control will display an asterisk (*) next to variable if the default value is changed. To set the variable back to default press the **[ORIGIN]** button.
6. Press the **[DOWN]** cursor arrow key to move to the next variable.
7. With all of the variables entered, you can press **[CYCLE START]** to immediately run the program in **MDI**, or **[F4]** to output the code to either the clipboard or **MDI** without running the program.

6.10 Y Axis

The Y Axis moves tools perpendicular to the spindle center line. This motion is achieved by compound motion of the X- and Y-Axis ballscrews.

Refer to G17 and G18, starting on page **275**, for programming information.

F6.16: Y-Axis Motion: [1] Y-Axis compound motion, [2] Horizontal plane.



6.10.1 Y-Axis Travel Envelopes

You can find detailed work and travel envelope information for your machine at data.haascnc.com/install.



TIP:

This site can be accessed through www.haascnc.com, by clicking on Owners> New Machine/ PRE-INSTALLATION.

Select your machine model, and then click on "Download Detailed Layout Drawings for the... PDF".

When you set up tooling for the Y Axis, consider these factors:

- Work piece diameter
- Tool extension (radial tools)
- Required Y-Axis travel from the centerline

6.10.2 Y-Axis Lathe with VDI Turret

The position of the work envelope shifts when using radial live tools. The length the cutting tool extends from the centerline of the tool pocket is the distance the envelope shifts.

You can find detailed work and travel envelope information for your machine at data.haascnc.com/install.



TIP:

This site can be accessed through www.haascnc.com, by clicking on Owners> New Machine/ PRE-INSTALLATION.

Select your machine model, and then click on "Download Detailed Layout Drawings for the.. PDF".

6.10.3 Operation and Programming

The Y Axis is an additional axis on the lathes (if so equipped) that can be commanded and behaves in the same manner as the standard X and Z Axes. There is no activation command necessary for Y Axis.

The lathe automatically returns the Y Axis to spindle centerline after a tool change. Make sure the turret is correctly positioned before commanding rotation.

Standard Haas G-codes and M-codes are available when programming with the Y Axis.

Mill type cutter compensation can be applied in both G17 and G19 planes when performing live tool operations. Cutter compensation rules must be followed to avoid unpredictable motion when applying and canceling the compensation. The Radius value of the Tool being used must be entered in the **RADIUS** column of the tool geometry page for that tool. The tool tip is assumed as "0" and no value should be entered.

Programming recommendations:

- Command Axes home or to a safe tool change location in rapids using G53 which moves all axes at the same rate simultaneously. Regardless of the positions of the Y Axis and X Axis in relation to each other, both move at the MAX possible speed towards commanded position and usually do not finish at the same time. For example:

```
G53 X0 (command for home) ;  
G53 X-2.0 (command for X to be 2" from home) ;  
G53 X0 Y0 (command for home) ;
```

Refer to G53 on page 282.

If commanding the Y and X Axes home using G28 the following conditions must be met and the described behavior expected:

- Address identification for G28:

X = U

Y = Y

Z = W

B = B

C = H

Example:

G28 U0 (U Zero) ; sends the X Axis to home position.

G28 U0 ; is okay with Y Axis below spindle centerline.

G28 U0 ; produces a 560 alarm if Y Axis is above spindle centerline. However homing the Y Axis first or utilizing a G28 without a letter address does not generate the 560 alarm.

G28 ; sequence sends X, Y, and B home first then C and Z

G28 U0 Y0 ; produces no alarm regardless of the Y-Axis position.

G28 Y0 ; is okay with Y Axis above spindle centerline.

G28 Y0 ; is okay with Y Axis below spindle centerline

Pressing **[POWER UP/RESTART]** or **[HOME G28]** produces the message:

Function locked.

- If X Axis is commanded home while the Y Axis is above spindle centerline (positive Y-Axis coordinates), alarm 560 is generated. Command Y Axis home first, then X Axis.
- If X Axis is commanded home and the Y Axis is below spindle centerline (negative Y-Axis coordinates), the X Axis goes home and Y does not move.
- If both X Axis and Y Axis are commanded home using G28 U0 Y0, the X Axis and Y Axis go home at the same time regardless of Y being above or below the centerline.
- Clamp the main and/or secondary spindles (if so equipped) anytime live tooling operations are being performed and C Axis is not being interpolated.



NOTE:

The brake unclamps automatically any time C-Axis motion for positioning is commanded.

- These canned cycles can be used with the Y Axis. Refer to page **262** for more information.

Axial Only Cycles:

- Drilling: G74, G81, G82, G83,
- Boring: G85, G89,
- Tapping: G95, G186,

Radial Only Cycles:

- Drilling: G75 (a grooving cycle), G241, G242, G243,
- Boring: G245, G246, G247, G248
- Tapping: G195, G196

Program Example of Y-Axis Milling:

F6.17: Y-axis Milling Program Example: [1] Feed, [2] Rapid.

